

Let $\mathbb{P}^2$ be a projective plane with homogeneous coordinates $[x : y : z]$, and let γ be the involution of $\mathbb{P}^2$ given by

$$[x : y : z] \mapsto [x : y : -z].$$

Then a plane sextic is γ -invariant if and only if it is defined by an equation of the form

$$f_6(x, y) + f_4(x, y)z^2 + f_2(x, y)z^4 + f_0(x, y)z^6 = 0,$$

where $f_\nu(x, y)$ is a homogeneous polynomial of degree ν . Hence the family $\mathcal{F}$ of γ -invariant sextics is parametrized by a 15-dimensional projective space. Note that a general member $C \subset \mathbb{P}^2$ of $\mathcal{F}$ is non-singular.

Proposition 0.1. *Let C be a general member of $\mathcal{F}$, and let $X_C \rightarrow \mathbb{P}^2$ be the double plane branching along C .*

- (1) *The plane sextic C has exactly 120 six-tangential conics.*
- (2) *The K3 surface X_C has exactly 240 smooth rational curves, which are pull-backs of the six-tangential conics of C .*
- (3) *The Picard lattice of X_C is of rank 9 with a Gram matrix*

$$\text{diag}(2, -2, -2, -2, -2, -2, -2, -2, -2).$$

The nef-and-big cone of X_C is a polyhedron with finitely many faces, and these faces are defined by the 240 smooth rational curves.

Proof.

□